

NAME

bdd – Bit stream copy and modify

SYNOPSIS

bdd [OPTIONS]

DESCRIPTION

bdd modifies information in bit streams (non byte-aligned).

By default, bdd reads bytes from standard input and writes them to standard output. With options, the behaviour can be widely modified to handle inputs and outputs with accuracy of one bit.

bdd considers input and output as streams of bit units. A bit unit is a repeatable units of constant amount of bits. Those units can be in the middle of the stream and the input stream can be infinite (like /dev/zero).

Every unit is divided to bit-defined fields and fields are interpreted into a tuple (analogy of relational database). Tuples may have integers, floats and strings.

Tuple can be modified and packed again bit by bit into a unit. Units are written into the output bit stream.

There can be another input stream, merge stream. Units in the merge stream are output after every main input stream unit. Merge stream does not support division into fields.

The name bdd comes from binary-dd because it is able to do much of the operations anyone would need in dd to handle bit streams.

OPTIONS**Input and output file selection:**

The default is to use standard input and standard output.

--input-file=FILENAME

Selects input file to used as input instead pipe input. Can not be used with --input-zeros --input-ones, --input-random or --input-counter. --input-integers and --input-tuples take their input from input file. Filename can also be "-" to describe pipe input (default).

--output-file=FILENAME

Selects output file to used as output instead pipe output. Filename can also be "-" to describe pipe output (default). Outuput file must not exist, bdd does not overwrite an existing file.

Input Units, Skips, and Gaps

A unit is a constant bit amount that repeats in the input stream. All stream dimensions are measured strictly in bits:

--input-unit=BITS

Breaks input into active units of BITS. If input is less than an even amount of units, the most significant bits of the last unit are taken from the input file and the least significant bits are filled with zeroes. Input unit defaults to 8 bits, or to the remaining bits in --input-raw-unit after --input-offset. If --input-pattern is defined, the combined bit size of the --input-pattern defines the input unit. You must not define both --input-unit and --input-pattern.

--input-skip-bits=BITS

Skip BITS bits from the beginning of the input before reading the first unit. Supports 64-bit integer values (up to 18.4 Exabits / 2.3 Exabytes), human-friendly size suffixes, and arbitrary multiplication expressions using `*` or `x` (e.g. `1000000*24` to skip one million 24-bit units, `1000*1000*24`, or `1920x1080*24`): binary multiples (K, M, G, T, P, E or Ki, Mi, Gi, Ti, Pi, Ei, e.g. `10G = 10*1024^3` bits), decimal multiples (KB, MB, GB, TB, PB, EB), and byte-scaled multiples (e.g. `10GiB = 10*1024^3*8` bits, `100B = 800` bits). All numeric arguments (skips, counts, offsets, units) accept multiplication expressions.

--input-skip-units=UNITS

Skip UNITS units (or raw units, if `--input-raw-unit` is set) from the beginning of the input. Supports size suffixes (e.g. `10M`, `1G`). If both `skip-bits` and `skip-units` are defined, both skips are performed. `--input-skip-units` and `--input-skip-bits` automatically seek directly over skipped bytes in $O(1)$ time on regular files rather than reading and discarding data from disk (see `--no-seek`).

--input-gap=BITS

Number of bits skipped after reading each unit. When `--input-raw-unit` is set, defines the gap in bits between successive raw units (default: 0).

Raw Units & Containers

In structured binary files, active units often reside inside larger repeating containers, words, or frames (such as 8-bit bytes, 16-bit half-words, or 32-bit words):

--input-raw-unit=BITS

Defines the repeating container or frame size in bits. Within each raw unit, an active unit of `--input-unit` bits is extracted starting at `--input-offset` bits. The remainder of the raw unit (`post-gap = raw-unit - (offset + unit)`) is automatically skipped without requiring manual calculation. Extra padding between successive raw units can be specified with `--input-gap`.

--input-offset=BITS

Bit offset of the active unit within each raw unit (default: 0). Requires `--input-raw-unit` to be specified.

--input-assert-aligned

Gives error message if the input is not full units. Note that with `--count` you can cleanly stop reading non-aligned input.

--no-seek, --do-not-seek

Globally disable seeking on all inputs (both primary and merge streams), forcing streaming sequential reads. By default, **bdd** automatically attempts $O(1)$ filesystem seek whenever the input stream is a seekable regular file or file descriptor, seamlessly falling back to sequential stream reads if the input is a pipe, socket, or FIFO.

--input-no-seek, --no-input-seek

Disable seeking specifically on the primary input stream, forcing sequential streaming reads.

--input-use-seek, --input-seek

Explicitly enable seeking on primary input (this is already the default whenever seeking is possible). Maintained for backwards compatibility.

--input-reverse-bytes

Every time when a byte is read from input, bits are reversed. 0x01 becomes 0x80. Look --input-little-endian for more information.

--input-reverse-unit

Every time when a unit is read, bits are reversed. Look --input-little-endian for more information.

--input-little-endian

By default, the input is assumed to be in network byte order or in big-endian byte order. However, the natural byte order for most processors is little-endian byte order. 16-bit big-endian integer 1 in byte stream is 0x00 0x01 and little-endian 0x01 0x00. In some systems, even if integers are little-endian, floating point numbers and strings are in big-endian order.

Unit is translated as an integer assuming that the least significant bit is the first one. To implement this as generic as possible, this is actually a synonym for combination --input-reverse-bytes and --input-reverse-unit. If we read 16-bit integer 1, in little-endian input stream it is 0x01 0x00. With --input-reverse-bytes it turns to 0x80 0x00 and with --input-reverse-unit it turns to 0x00 0x01. By these two steps, the operation has a meaningful interpretation even in non-byte-aligned cases.

After --input-little-endian, the content is usually read as a big-endian number in --input-pattern!

Bypass bit stream input (fake inputs)

Instead of the file input, the input can be one of these fake inputs. You can use either file input or one of these fake inputs. You must use --count to tell how many units the fake input has (except for --input-integers).

Values of fake inputs are truncated to the input unit size. For example, if you use --input-counter with input unit size 8 and output unit size 9, the topmost bit of the output is always zero.

If a fake input is used, the standard input is not opened at all for input, and standard input can be used with --merge-file.

None of the fake inputs does support reversing bits, initial skips or gap operations. --input-random, --input-counter and --input-integers support --skip (also others allow it, but it does nothing as it would not affect the output).

--input-zeros

All the input units are zeros.

--input-ones

Like --input-zeros but all the bits of the input unit are ones.

--input-random

Like --input-zeros but all the input bits are random (from /dev/urandom).

--input-counter

Like --input-zeros, but input is units, every one with one bigger value than the previous. Counter value is cut to input unit size. Can not be reversed but --skip can be used to start from a specific number. If --skip=N, the first counter value will be N.

--input-integers

Instead of a bit stream, input is text, 10-base unsigned integers that are separated with newline characters. Whitespace around number is allowed and empty lines are skipped. Can not be used with any other unit selection argument but can be used with --skip and --count to read only a part of the file.

Unpack a unit to a tuple

The input unit is split into fields according the --input-pattern. Fields are possibly interpreted as typed values (signed integers, floats or strings).

If there is no --output-pattern, the first field of the tuple is output as the sole unit. It must be a positive integer to be interpreted as a unit. If there is no --input-pattern, --output-pattern can still be used but there is only one field available and it is the input unit as whole.

--input-pattern=PAT

Interpret every unit as fields as described in the PAT. PAT consists of patterns that match some bits in the input unit. A pattern has first the number of the bits matched and then a character that gives those bits the interpretation. For example,

--input-pattern=1U7U

divides a 8-bit unit into two integers, the first bit forms one field and seven next bits form another field. U means that the fields are interpreted as unsigned integers.

x - - skip these bits without output
 U - i - field is unsigned integer (msb first)
 u - i - field is unsigned integer (lsb first)
 M - ii - reads bits as 2's complement signed integer,
 but stores sign and magnitude separately as
 two positive integer fields
 m - ii - same as M, lsb first

M and m generate two fields from one number. The sign is output first as one integer and the magnitude second as another integer. The sign is zero for positive numbers and one for negative numbers. Note that magnitude alone needs as many bits as the original number (look --cut-maxint).

The following type characters are only for rare, advanced use. The main use is that they can give a human interpretation of the input stream using --output-tuples.

If the need is only rearrange bits, you should use only U pattern. The following type characters give some functionality that is sometimes necessary and it makes sense to output them to the bit stream:

S - i - field is 2's complement signed integer (msb)
 s - i - field is 2's complement signed integer (lsb)
 F - d - IEEE-754 32-bit floating point number
 f - d - same, lsb first
 D - d - IEEE-754 64-bit floating point number
 d - d - same, lsb first
 16H - d - IEEE-754 16-bit half precision float (FP16, msb)
 16h - d - IEEE-754 16-bit half precision float (FP16, lsb)
 16Y - d - Google Brain 16-bit bfloat16 (BF16, msb)
 16y - d - Google Brain 16-bit bfloat16 (BF16, lsb)

```

8E - d - OCP FP8 E4M3FN (8-bit AI float, msb)
8e - d - OCP FP8 E4M3FN (8-bit AI float, lsb)
8Q - d - OCP FP8 E5M2 (8-bit AI float, msb)
8q - d - OCP FP8 E5M2 (8-bit AI float, lsb)
6E - d - OCP FP6 E3M2 (6-bit AI float, msb)
6e - d - OCP FP6 E3M2 (6-bit AI float, lsb)
4E - d - NVIDIA Blackwell / OCP FP4 E2M1 (4-bit AI float, msb)
4e - d - NVIDIA Blackwell / OCP FP4 E2M1 (4-bit AI float, lsb)
c - s - field is unicode UTF-8 character string
C - s - same, lsb first

```

Patterns also support repetition multipliers in the format *COUNT*SPEC* or *COUNT*(SPEC1 SPEC2 ...)*. For example, *4*8B* expands to four 8-bit unsigned bytes, and *2*(4U4u)* expands to *4U4u4U4u*.

S and s should be used with caution. If negative value is used in making a unit, it will cause an error. The only way to use negative values correctly is *--output-pattern=nS*, *--abs* or *--output-tuples*.

Floating-point numbers can be read and written using any of the supported float patterns: *64D/64d* (FP64), *32F/32f* (FP32), *16H/16h* (IEEE FP16), *16Y/16y* (Google Brain BF16), *8E/8e* (OCP FP8 E4M3FN), *8Q/8q* (OCP FP8 E5M2), *6E/6e* (OCP FP6 E3M2), and *4E/4e* (NVIDIA Blackwell / OCP FP4 E2M1).

Internally, all input floats are decoded into IEEE 754 64-bit doubles (*f64*) in the tuple field pipeline. When written out through an output pattern, *bdd* encodes the internal *f64* value directly into the requested output float format using standard round-to-nearest-even rules (or round-to-nearest). This provides native, loss-free upcasting and bit-exact downcasting/quantization between arbitrary floating-point representations (e.g. *--input-pattern=32F --output-pattern=16H* or *--input-pattern=16H --output-pattern=4E*). The *--round* manipulator can additionally be used to apply explicit rounding modes (such as **floor**, **ceil**, **trunc**, or **round_ties_even**) prior to packing.

Most likely, you want to use upper case variants of the patterns (bitwise big-endian representation). If you want to interpret bitwise little-endian numbers, you likely want to use *--input-little-endian* and still use uppercase patterns. Lowercase patterns are for the (rare) case that the value in bit stream is exactly in the bitwise (not bitwise) opposite order, least significant bit first.

Rule to remember: uppercase letters are big-endian, lowercase letters are little-endian.

Floating point number length must match the bit length. D must be always written as pattern *64D*.

Also, the string must be right amount of bytes. Some byte strings can not be output as a valid UTF-8 string.

Bypass unit and field selection

--input-tuples

Instead of a bit stream, input is a text file containing tuples. Tuple is a line of comma-separated values like:

```
123, -55, 678.5, "abba"
```

Tuples are separated with newline. Whitespace around the comma is allowed. If *--input-tuples* is used, unit or field selection arguments have no effect but *--count*, *--skip* and tuple manipulation options can be used.

Allowed types and their internal representations are:

```

Integers (signed bignums)
Floats (translated as IEEE 754 64-bit floating point numbers)

```

Strings (UTF-8)

Manipulate tuple

Fields are represented with field numbers (starting from 0, negative indices count from the end with -1 being the last field). The manipulation pipeline is ordered and stackable: operations execute in the exact order they are specified on the command line. Multiple operations (such as arithmetic, bit shifts, logical masks, filtering, and reordering) can be chained together.

--rearrange=FIELDLIST

Selects fields from input tuple for output tuple. Comma-separated list of field numbers starting from 0 (field 0 is the first field in the tuple). Negative field number refers to the field counting from the end of the tuple (-1 is the last field). The same field number can exist several times duplicating the input field. Not all input fields need to be in the output tuple.

--add=FIELD,VALUE

Adds VALUE to the numerical value of FIELD.

--sub=FIELD,VALUE

Subtracts VALUE from the numerical value of FIELD.

--mul=FIELD,VALUE

Multiplies FIELD by VALUE.

--div=FIELD,VALUE

Performs integer or floating-point division of FIELD by VALUE.

--mod=FIELD,VALUE

Computes modulo remainder of FIELD by VALUE.

--shift-left=FIELD,BITS

Shifts FIELD left by BITS bits (multiplies by 2^{BITS}).

--shift-right=FIELD,BITS

Shifts FIELD right by BITS bits (integer division by 2^{BITS}).

--and=FIELD,MASK

Performs bitwise AND of FIELD with MASK (decimal, 0x hex, or 0b binary).

--or=FIELD,MASK

Performs bitwise OR of FIELD with MASK (decimal, 0x hex, or 0b binary).

--xor=FIELD,BITS

Inverts the rightmost BITS bits of FIELD, or applies a direct bitwise XOR mask if specified in hex or binary format. For example, the sign of a number read using pattern "M" can be changed using **--xor=0,1**.

--not=FIELD

Inverts all bits of FIELD.

`--abs=FIELD`

Removes sign from the value of the field (calculates absolute value). Legacy syntax `--abs=FIELD,0` is also accepted.

`--sign=FIELD`

Extracts the sign of the field: 1 if the value was negative and 0 if non-negative. Legacy syntax `--sign=FIELD,0` is also accepted.

`--filter=FIELD,OP,VALUE`

Filters tuples based on a predicate condition where OP is one of `==`, `!=`, `<`, `<=`, `>`, or `>=`. Tuples failing the predicate are discarded from the output stream.

`--round=FIELD,LIMIT[,MODE] | --round=FIELD,MODE`

Rounds floating-point fields or bounds/clamps integer fields within `[-LIMIT, LIMIT]` (or `[0, LIMIT]` if unsigned). Visible alias: `--cut-maxint`. When LIMIT is omitted (e.g. `--round=0, floor` or `--round=0, round_ties_even`), the rounding mode is applied directly without magnitude clamping. An optional MODE parameter may be appended using comma or colon delimiter (e.g. `--round=0, 127, wrap` or `--round=0, 127:round_ties_even`). Supported modes:

- **saturate** or **clamp** (default): Clamps out-of-range values to MAXINT or -MAXINT.
- **wrap** or **wrapping**: Wraps values around using modular arithmetic (modulo MAXINT+1 for unsigned, or across `[-MAXINT, MAXINT]` for signed).
- **zero** or **reset**: Replaces out-of-range values with zero.
- **drop** or **filter**: Discards the entire tuple from the output stream if value exceeds bounds.
- **trunc**: Truncates magnitude toward zero.
- **floor**: Rounds toward negative infinity.
- **ceil**: Rounds toward positive infinity.
- **round**: Rounds to nearest neighbor, ties away from zero.
- **round_ties_even**: Rounds to nearest neighbor, ties to even digit (Rust and IEEE 754 default banker's rounding).

Field number can be negative when the position is counted from the end of the tuple.

`--remove-right=FIELD,BITS`

If we want to output other bits from the field than the rightmost ones, we can remove BITS rightmost bits from the field.

Skip a part of tuples

`--skip=N`

N first units are skipped.

`--count=N`

Only N units are processed. For example, if `--skip=1` and `--count=1`, only the second unit is processed.

Pack a tuple to a unit

`--output-pattern='PAT'`

Packs output tuple to output bit unit. Fields must be in order, use `--rearrange` to arrange fields to the order of output pattern. Types are the same as the ones with the input pattern with the

following exceptions:

- z - fill with zero
- o - fill with one
- r - fill with random (from /dev/urandom)
- x - can not be used in the output pattern

Also in the output-pattern, lowercase letters turn the bit order reversed. Look --input-pattern for more information.

z, o and r do not consume fields from the tuple. m and M consume two fields from the tuple and convert them to one 2's complement bit unit.

If signed value is too big to a signed field, the magnitude is first truncated to the right size. 4-bit signed output of -17 is '1111', because 17 is first truncated to 4 bits giving '0001' or 1 and the bit representation of -1 is '1111'.

If tuples are used (--input-pattern or --input-tuples), but there is no --output-pattern option, the first field in the tuple is used directly as the output unit. The first field of the tuple must be unsigned integer if it is used as an output unit.

Collect output data stream

--output-unit=BITS

Packs the output unit to BITS bits. If --output-pattern is defined, output unit size comes from the total size of the output pattern and --output-unit must not be defined. If there is neither --output-unit nor --output-pattern, the default output unit size is 8 bits.

The last output byte is filled with zeros, if needed.

--output-little-endian

Unit is written to the output stream least significant byte first. The default is the network byte order or big-endian byte order where the most significant byte is written first. Actually --output-little-endian is a combination of --output-reverse-unit and --output-reverse-bytes and as such, those options can not be used with --output-little-endian. Look --input-little-endian for more discussion.

--output-reverse-unit

Reverses bits in the output unit. Look --output-little-endian for more information.

--output-reverse-bytes

Reverses bits of every written byte. Look --output-little-endian for more information.

Merge output with another data stream:

Units of the merge file are written to the output after every normal output unit. Output is finished when the normal output is finished. Last merge unit is written after the last output unit so that as many merge units as normal units are written. It is an error if the merge file does not have enough units but it can contain more units than the normal input file and the extra units are ignored.

Merge unit is defined like the normal input unit. Merge unit is written to the output with the same unit size as it has in input. Merge unit size can be different than the normal output unit. Output of the merge unit is done with the same --reverse-bytes and --reverse-unit options as the normal output.

--merge-file=FILENAME

Opens the merge file. If FILENAME is "-", mergefile is standard input. If mergefile is standard input, input file must not be the same.

--merge-unit=BITS

Breaks merge input to units of BITS.

--merge-skip-bits=BITS

Skip BITS bits from the beginning of the merge input

--merge-skip-units=UNITS

Skip UNITS units from the beginning of the merge input.

--merge-copy-first=BITS

Copy BITS bits as-is from the beginning of the merge file before writing the first unit. If both skip (bits and/or units) and --merge-copy-first are defined, skip is done first.

--merge-gap=BITS

When --merge-raw-unit is set, defines the gap in bits between successive merge raw units (default: 0). When --merge-raw-unit is omitted, specifies the bits skipped after writing each unit.

--merge-raw-unit=BITS

Defines the repeating container or frame size in bits for the merge stream. Within each merge raw unit, an active unit of --merge-unit bits is extracted starting at --merge-offset bits. The remainder of the merge raw unit is automatically skipped.

--merge-offset=BITS

Bit offset of the active unit within each merge raw unit (default: 0). Requires --merge-raw-unit to be specified.

--merge-assert-aligned

Warn if the whole merge input was not used or the merge input did not end to the byte border.

--merge-no-seek, --no-merge-seek

Disable seeking specifically on the merge input stream, forcing sequential streaming reads. By default, seeking is automatically attempted whenever the merge source is a seekable regular file.

--merge-use-seek, --merge-seek

Explicitly enable seeking on merge input (this is already the default whenever seeking is possible). Maintained for backwards compatibility.

--merge-little-endian

Merge file is read with both --merge-reverse-bytes and --merge-reverse-unit on.

--merge-reverse-bytes

Every byte read from merge file is reversed.

`--merge-reverse-unit`
Every unit read from merge file is reversed.

Show output as human readable

`--output-tuples`
Replace bit output with human-readable output of tuples. This output is readable with `--input-tuples`

`--output-integers`
Replace bit output with human-readable output of units as integers. This output is readable with `--input-integers`

If merge file is used, the main unit and the merge unit is printed separately as two integers (possibly with different unit size).

The integer is truncated to the unit size but reversing bits does not affect to the integer output.

`--output-hex`
Replace bit output with human-readable output of units as hex numbers.

Units are written as they would be written to the bit stream except that `--output-little-endian`, `--output-reverse-bytes` and `--output-reverse-unit` do not have effect to the hex output.

Output unit size affects the number as it affects in the bit output. If output unit size is 3 bits, the number 8 is printed as hex 0. The hex number has as many digits as needed to represent the full unit but topmost zero bits are not necessarily a part of the output unit. Units from the merge file are written with their own output unit so that they may have different amount of digits.

Output is divided to lines so that in one line there is 1, 2, 4, 8, 16 or 32 units separated with spaces. The line length is always between 40 and 80 characters.

`--output-bits`
Replace bit output with human-readable output of units as bits. Look `--output-hex` for more information.

`--output-json`
Output stream of structured JSON objects (NDJSON), one object per line with typed field names and decoded values.

`--output-csv`
Output stream as comma-separated values (CSV).

`--csv-header=COL1,COL2,...`
Specify custom column names for CSV output header row.

`--output-visual`
ANSI color-coded visual stream inspector highlighting bit boundaries and field values.

Stream demuxing and routing**--demux=FIELD:PATH**

Extract individual tuple field by index and stream it directly to a dedicated file or FIFO. Can be repeated.

--demux-files=PATH1,PATH2,...

Demux tuple fields in order to a comma-separated list of target files.

--input-repeat=COUNT

Loop through the input stream COUNT times (supports seeking or memory buffering).

EXAMPLES**bdd < foo > bar**

Just copies file foo to bar, one byte at time.

bdd --input-unit=3

Extracts 3-bit stream values from standard input and expands values to bytes in standard output. If the file is not aligned to 3 bits, the missing least significant bits for the last unit are zeros.

bdd --input-unit=1 --output-pattern=1U1z

Output stream is input stream with zero added after every bit. Four two-bit units are packed to one byte.

bdd --input-pattern=8U8U --rearrange=1,0 --output-pattern=8U8U

Swap two consecutive bytes.

echo -en '1,2,30,7,7' | bdd --input-tuples --output-pattern='2U3U3U' | od -t o1

Build octal numbers from fields (prints 123 377)

bdd --input-counter --count=16 --output-hex

Prints:

```
00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f
```

bdd --input-counter --count=16 --input-unit=12 --output-unit=12 --merge-file=/etc/passwd --output-hex

Prints first 16 bytes of /etc/passwd file as hex preceeded by a three-digit hex address of the byte:

```
000 72 001 6f 002 6f 003 74 004 3a 005 78 006 3a 007 30
008 3a 009 30 00a 3a 00b 72 00c 6f 00d 6f 00e 74 00f 3a
```

SEE ALSO

dd(1)

BUGS

None known in the current release. Please report issues at <https://github.com/turtiainen/bdd>.

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